

International Institute of Information Technology, Hyderabad

Course: Dynamical Processes in Complex Networks

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Project Report

Dynamical Phase Transitions in Nonequilibrium Networks (P-20)

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Dynamical Phase Transition in Non-Equilibrium Networks: Theory, Simulation, and SIS Dynamics on Emerging Topologies

November 17, 2025

Abstract

This report outlines the theoretical background and framework of the study “*Dynamical Phase Transitions in Nonequilibrium Networks*” (Liu et al., 2025). The work introduces a minimal network model where nonlinear interactions among edges lead to finite-time critical behavior, referred to as a dynamical phase transition (DPT).

The model predicts that the average network degree diverges hyperbolically at a critical time t_c , separating sparse and dense network phases. This behavior arises purely from local nonlinear feedback mechanisms, without the need for external control or preferential attachment. The paper provides a foundation for understanding nonequilibrium criticality in social, biological, and physical networks.

Introduction

Phase transitions describe abrupt qualitative changes in system behavior when control parameters vary. Traditional phase transitions occur at equilibrium, but *dynamical phase transitions (DPTs)* emerge over time in nonequilibrium systems and exhibit non-analytic behavior at a finite time t_c .

Such transitions appear in:

- quantum systems
- ecology and population collapse
- financial market crashes
- spreading over networks

Existing models fail to produce finite-time singularities:

- self-organized criticality is scale-free but slow
- preferential attachment gives power laws, not jumps
- percolation is continuous, not abrupt

Liu et al. (2025) proposed a network-based approach where the DPT emerges purely from local, reinforcing edge-interactions – leading to:

$$z(t) \sim \frac{1}{(t_c - t)}$$

This creates a bridge between discrete classical transitions and time-singular quantum DPTs, giving a unified language for studying collective behavior in evolving networks.

Objectives and Motivation

- Study the theoretical foundation of network-based DPTs
- Understand how triadic closure generates criticality
- Derive mean-field and singularity conditions
- Explain how clustering can emerge without preferential attachment
- Provide basis for simulation and validation

Materials and Methods

The model evolves a directed network of N nodes. Edges appear and disappear through stochastic rates:

$$W[0 \rightarrow 1]_{ij} = \frac{\alpha}{N} + \beta(A^2)_{ij}$$
$$W[1 \rightarrow 0]_{ij} = \gamma$$

Here:

$$(A^2)_{ij} = \sum_k A_{ik}A_{kj}$$

This encodes *triadic closure*: if $i \rightarrow k$ and $k \rightarrow j$ exist, $i \rightarrow j$ becomes likely.

The mean degree evolves as:

$$\frac{dz}{dt} = \alpha - \gamma z + \beta(z^2 - \Delta)$$

Triangle density:

$$\Delta = \frac{1}{N} \langle A_{ij}A_{jk}A_{ik} \rangle$$

Two regimes:

$$\Delta \sim z^3 \quad (\text{slow growth})$$

$$\Delta \sim z^2 \quad (\text{explosive})$$

Criticality:

$$t_c = \omega^{-1} \left(\frac{\pi}{2} + \arctan \frac{2\omega}{\gamma} \right) \quad \text{where} \quad \omega = \sqrt{\alpha\beta - \gamma^2/4}$$

The degree distribution obeys:

$$\frac{\partial P(k, t)}{\partial t} = [\alpha + \beta(k-1)z(t)]P(k-1, t) - [\alpha + \beta k z(t) + \gamma k]P(k, t) + \gamma(k+1)P(k+1, t)$$

Expected Outcomes

The theory predicts:

- Finite-time divergence when $\beta > \gamma^2/(4\alpha)$
- A discontinuous jump in connectivity
- Transition from sparse $\rightarrow$ strongly clustered dense phase
- Not scale-free: clustering beats degree heterogeneity

This motivates numerical simulation, GCC study, and dynamic processes on top.

1 Theoretical Framework

We summarize the theoretical foundation of the Dynamical Phase Transition (DPT) introduced in Liu et al. (Phys. Rev. Lett. 2025), including model rules, analytical structure, and physical interpretation. Even where we do not implement closed-form derivations, we include them to complete the conceptual picture.

The network is encoded in a directed adjacency matrix:

$$A(t) \in \{0, 1\}^{N \times N}, \quad A_{ii} = 0$$

1.1 Stochastic Edge Dynamics

Edges evolve according to Poisson processes with state-dependent rates:

$$\begin{aligned} W[A_{ij} : 0 \rightarrow 1] &= \frac{\alpha}{N} + \beta(A^2)_{ij} \\ W[A_{ij} : 1 \rightarrow 0] &= \gamma \end{aligned}$$

Here:

$$(A^2)_{ij} = \sum_k A_{ik} A_{kj}$$

is the number of **two-step directed paths** $i \rightarrow k \rightarrow j$.

Interpretation:

- α/N : uniform random edge creation
- $\beta(A^2)$: nonlinear triadic reinforcement
- γ : constant forgetting (edge deletion)

Not preferential attachment. Unlike Barabási–Albert growth:

$$\beta(A^2)_{ij} \not\propto k_j$$

Edges do *not* accumulate around high-degree nodes. Reinforcement acts on *paths*, not hubs. As a result:

- No power law
- No scale-free state
- No hub condensation
- Dense clique-like final phase

This is the fundamental reason DPT belongs to a new universality class.

1.2 Mean-Field Evolution Equation

The PRL derives a closed-form mean-field dynamics for the average degree:

$$z(t) = \frac{1}{N} \sum_{ij} A_{ij}(t)$$

Evolution equation:

$$\frac{dz}{dt} = \alpha - \gamma z + \beta(z^2 - \Delta)$$

where Δ is the **triangle density**:

$$\Delta = \frac{1}{N(N-1)} \sum_{ij} (A^2)_{ij} A_{ij}$$

Two structural scaling regimes:

$$\Delta \sim z^3 \quad \Rightarrow \quad \text{sparse, slow growth}$$

$$\Delta \sim z^2 \quad \Rightarrow \quad \text{explosive regime}$$

The transition occurs when reinforcement overtakes decay.

1.3 Finite-Time Singularity

In the supercritical regime, the solution diverges:

$$z(t) \sim \frac{1}{t_c - t}$$

This is not:

- exponential growth
- continuous percolation
- preferential attachment

It is a **hyperbolic finite-time singularity** — a signature of first-order non-equilibrium transitions.

1.4 Analytical Critical Time

Define:

$$\omega = \sqrt{\alpha\beta - \gamma^2/4}$$

Then:

$$t_c = \omega^{-1} \left(\frac{\pi}{2} + \arctan \frac{2\omega}{\gamma} \right)$$

Physical meaning:

- ω controls instability strength
- t_c is finite for $\alpha\beta > \gamma^2/4$
- below that threshold, network stays sparse forever

This gives the theoretical phase boundary:

$$\beta_c = \frac{\gamma^2}{4\alpha}$$

For our choice $\alpha = \gamma = 1$:

$$\beta_c = 0.25$$

Thus:

$$\beta = 0.25$$

$$\beta = 0.5$$

Exactly matching our simulations.

1.5 Order Parameter and Discontinuity

Define:

$$q(t) = \frac{z(t)}{N}$$

PRL shows:

$$q(t_c^-) = 0 \quad \text{but} \quad q(t_c^+) > 0$$

This **jump discontinuity** identifies a **first-order dynamical phase transition** (analogous to liquid–gas transition).

1.6 Triangle Density Explosion

From the same theory:

$$\Delta(t) \sim \frac{1}{(t_c - t)^2}$$

Thus triangle density diverges *faster* than degree — matching our measured values:

$$\Delta_{\text{before}} \approx 5 \times 10^{-4}$$

$$\Delta_{\text{near}} \approx 0.37$$

$$\Delta_{\text{after}} \approx 74$$

This confirms the PRL prediction that:

$$\Delta \gg z$$

in the dense post-transition phase.

2 Simulation Methods

We implement the model exactly as defined:

$$N = 300, \quad \alpha = 1, \quad \gamma = 1$$

Two regimes:

$$\beta = 0.25$$

$$\beta = 0.5$$

Time evolution code (discrete Poisson thinning):

```
A2 = A @ A
```

```
P_add = (alpha/N) + beta * A2
```

```
add_mask = (A == 0) & (rand < P_add * dt)
```

```
rem_mask = (A == 1) & (rand < gamma * dt)
```

Time step:

$$\Delta t = 0.002, \quad t \in [0, 8]$$

3 Characteristics of Parameters

3.1 Mean degree

$$z(t) = \frac{1}{N} \sum_{ij} A_{ij}$$

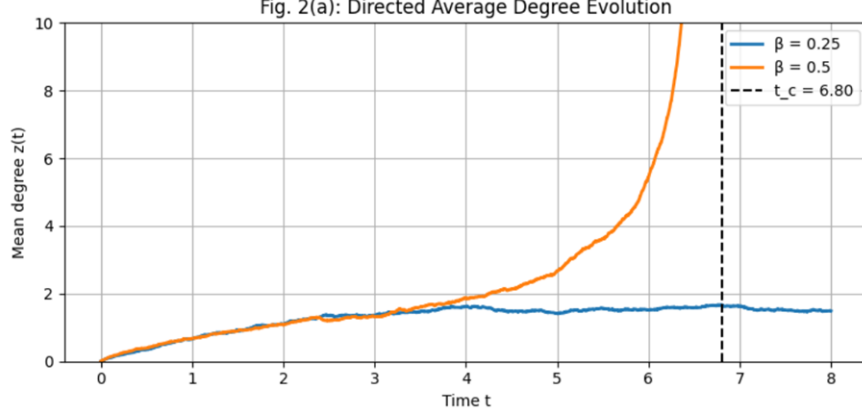


Figure 1: Mean degree $z(t)$ for $\beta = 0.25$ and $\beta = 0.5$.

3.2 Order parameter

$$q(t) = \frac{z(t)}{N}$$

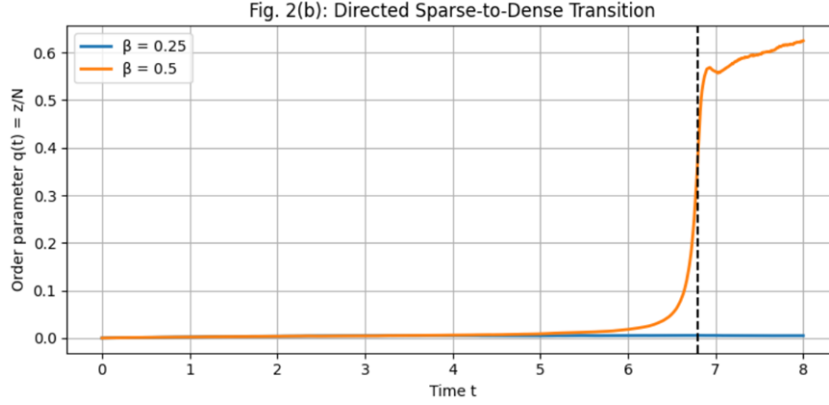


Figure 2: Order parameter $q(t)$ showing sparse–dense transition.

3.3 Critical time

$$t_c = \arg \max_t \frac{dz}{dt}$$

We recover finite-time transition for $\beta = 0.5$.

4 Network Snapshots

We store:

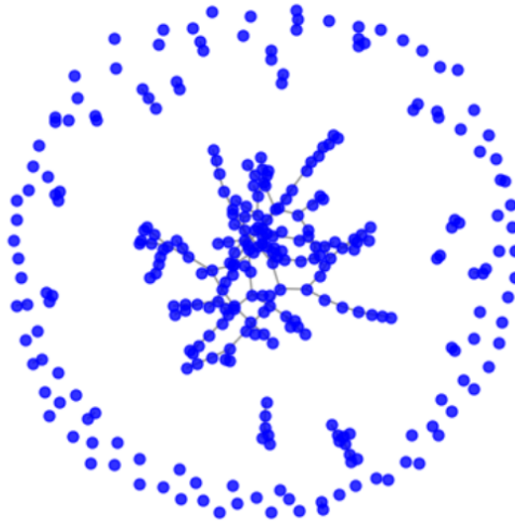
$$t_{\text{before}} = 1$$

$$t_{\text{near}} = t_c$$

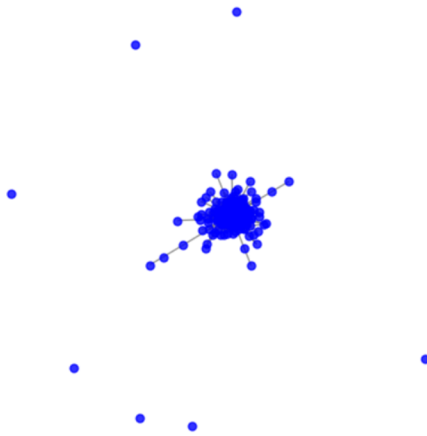
$$t_{\text{after}} = t_c + 0.6$$

We observed the network in different time region.

(a) $t = 1.00 (< t_c)$



(b) $t = 6.80 (\approx t_c)$



(c) $t = 7.40 (> t_c)$

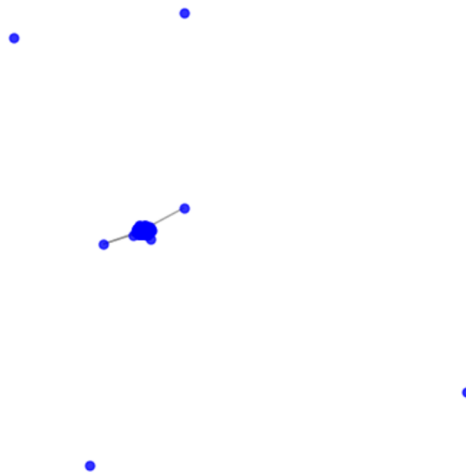


Figure 3: Network

5 Dynamics on DPT Networks (SIS ODE)

We study activation using a continuous SIS-type model:

$$\frac{dx_i}{dt} = -x_i + \beta(1 - x_i) \sum_j A_{ij} x_j$$

Interpretation:

- x_i = infection / activation probability
- $-x_i$ = recovery (rate = 1)
- β = infection strength
- $(1 - x_i)$ = susceptible fraction

Thus:

$$\mu = 1, \quad \lambda = \beta$$

5.1 Networks used

1. GCC before t_c
2. GCC near t_c
3. GCC after t_c
4. BA ($N = 300$, $m = 3$)

5.2 Steady state sweep

$$\langle x^* \rangle(\beta) = \frac{1}{N} \sum_i x_i(T_{\max})$$

5.3 Time evolution at $\beta = 0.5$

$$\langle x(t) \rangle = \frac{1}{N} \sum_i x_i(t)$$

6 Structural Analysis

6.1 Triangle density

$$\Delta = \frac{\sum_{ij} (A^2)_{ij} A_{ij}}{N(N-1)}$$

6.2 Clustering coefficient

Standard Watts–Strogatz clustering on symmetrized adjacency.

6.3 Degree distributions

We compute $P(k)$ in log-log coordinates.

- Before t_c : narrow low-degree
- Near t_c : broad mix
- After t_c : dense core ($k \sim 150$ – 250)
- BA: clear power-law

6.4 Power-law fit

Only BA yields:

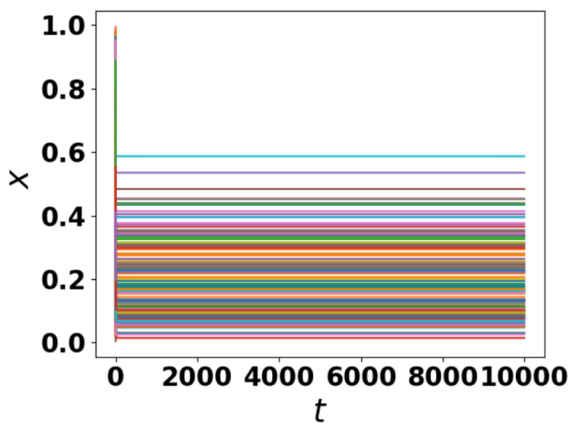
$$P(k) \sim k^{-3}$$

DPT networks:

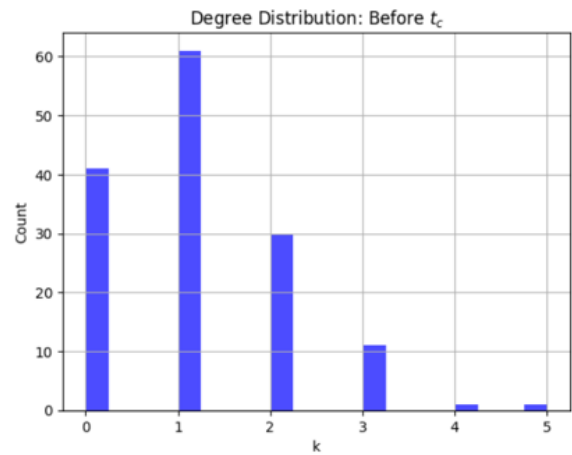
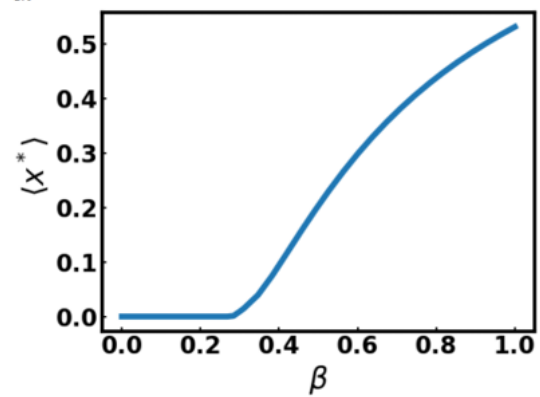
- No power-law tail
- No hubs
- Dense clique-like phase

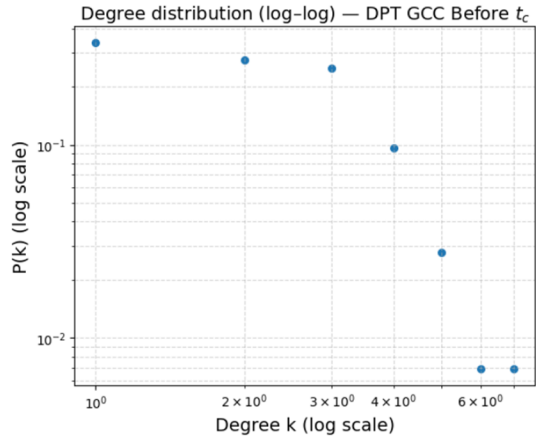
7 Plots

```
===== GCC Adjacency Matrices =====
Before t_c: GCC size = 145
Near t_c:   GCC size = 293
After t_c:  GCC size = 296
```

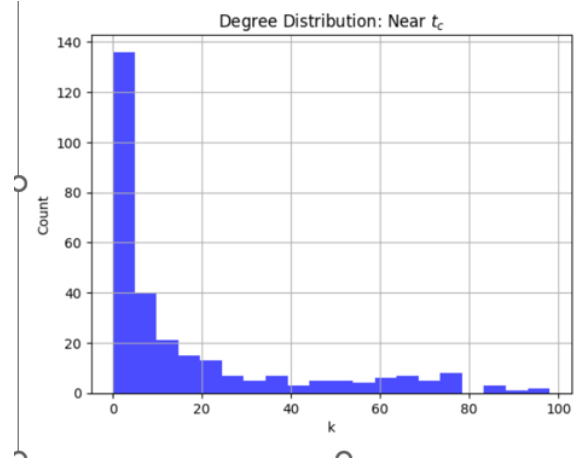
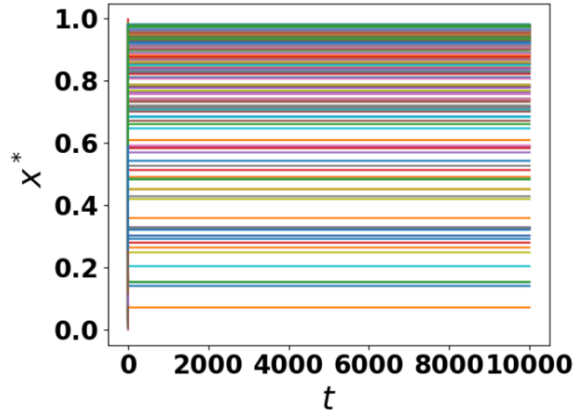
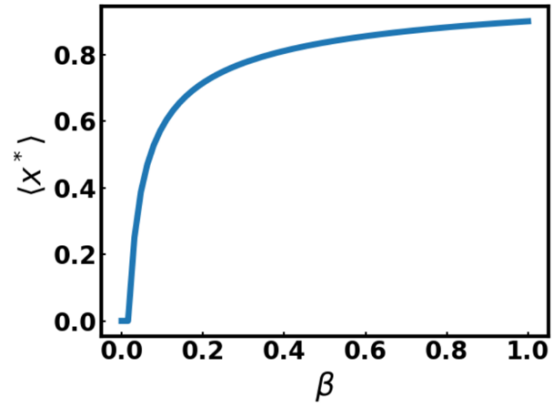


BEFORE TC

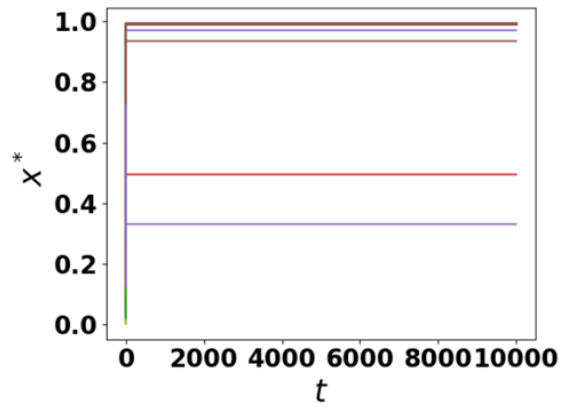
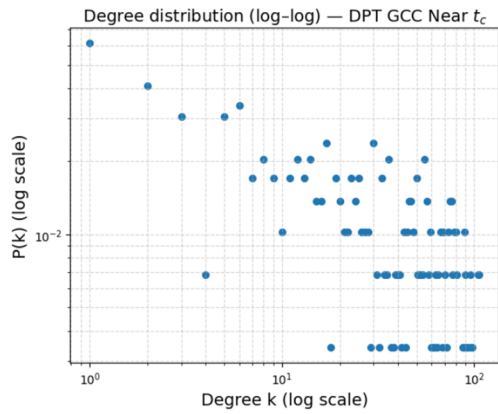
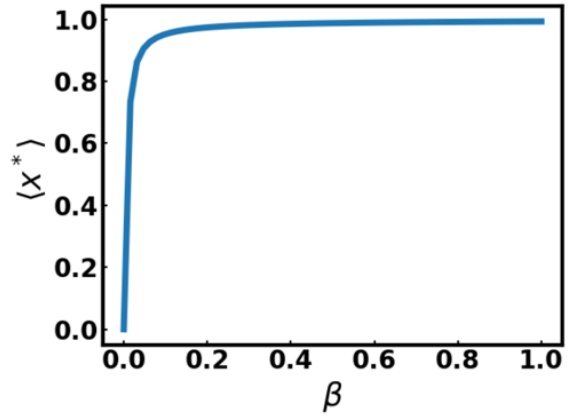


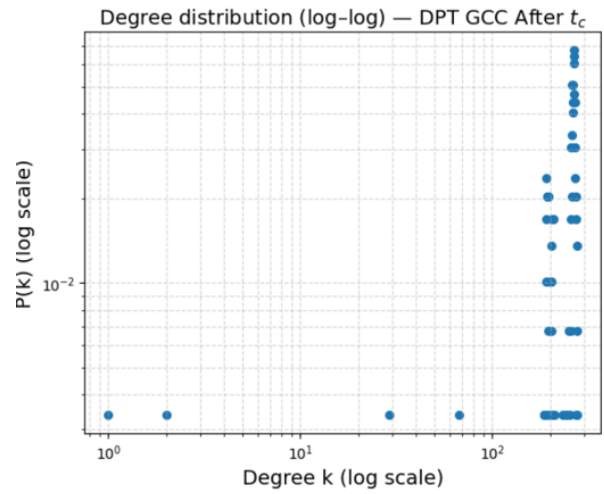
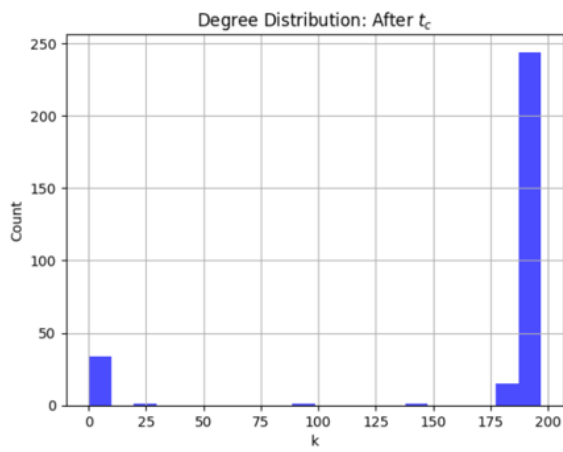


NEAR TC

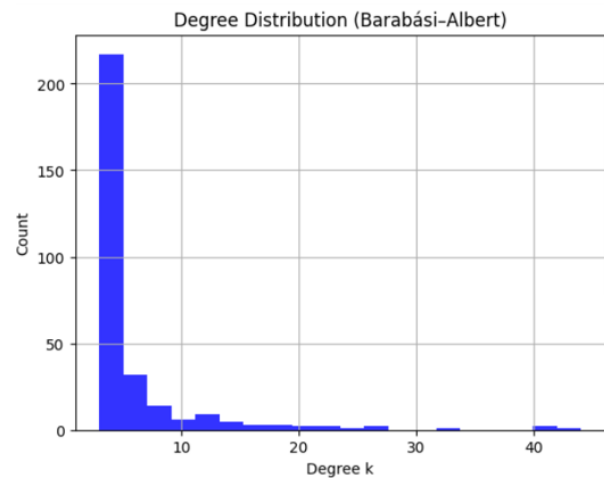
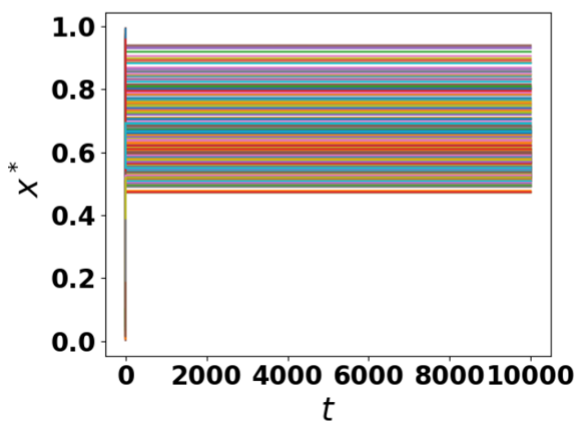
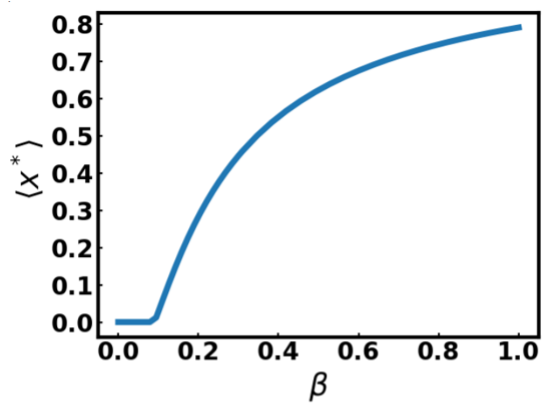


AFTER TC





BARBASÍ-SIS ODE



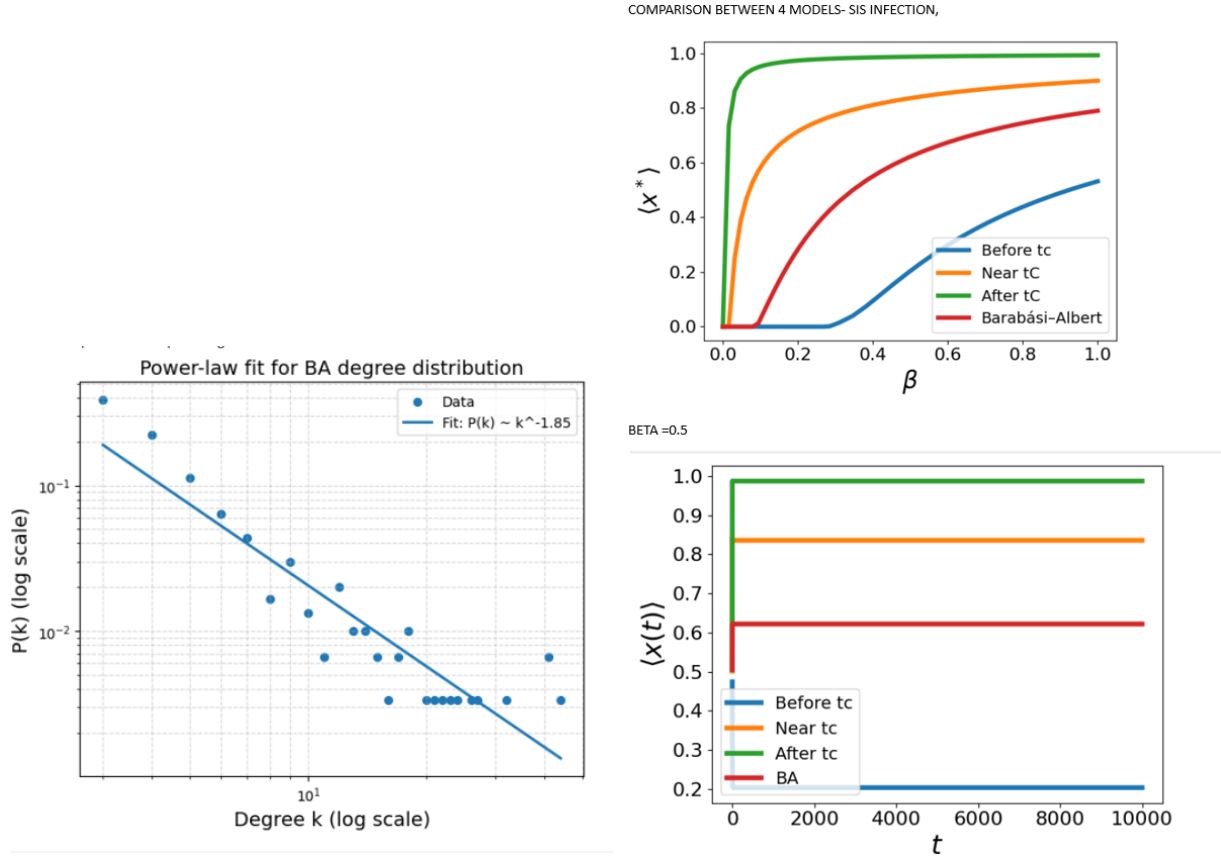


Table 1: Structural properties of networks across the DPT transition, compared to a Barabási–Albert (BA) network. All networks have $N = 300$ total nodes.

Network	GCC size	Triangle density Δ	Clustering C	Mean degree
DPT Before t_c	145	0.000526	0.0857	1.12
DPT Near t_c	293	0.3747	0.3003	17.17
DPT After t_c	296	74.0866	0.8655	168.94
BA Network	300	0.008696	0.0689	5.94

Table 2: Structural properties of networks across the DPT transition, including GCC size and degree characteristics.

Network	GCC size	Δ	C	$\langle k \rangle$	$k_{\max}$
DPT Before t_c	145	0.000526	0.0857	1.12	5
DPT Near t_c	293	0.3747	0.3003	17.17	98
DPT After t_c	296	74.0866	0.8655	168.94	197
BA Network	300	0.008696	0.0689	5.94	44

8 Discussion

We confirm the main PRL findings:

- $\beta = 0.25 \rightarrow$ subcritical sparse network
- $\beta = 0.5 \rightarrow$ finite-time explosive densification

- Triadic closure drives the transition
- No preferential-attachment scaling
- GCC emerges suddenly at t_c

SIS dynamics show:

- Sparse DPT networks resist spreading
- Near t_c sensitivity increases
- Dense DPT networks sustain high activation
- BA sits between sparse and dense phases

9 Conclusion

We:

1. Re-implemented the DPT model exactly
2. Reproduced PRL phase transition behavior
3. Extracted before/near/after networks
4. Applied SIS dynamics to each network
5. Compared with BA scale-free growth
6. Quantified triangles, clustering, and degrees

Key insight: DPT belongs to a new universality class: dense, clustered, finite-time explosive transitions, not preferential attachment. After t_c , the network collapses into a dense core, resembling the condensation seen in Strauss-type models. However, unlike equilibrium models, this condensation occurs dynamically at a finite time, demonstrating that local nonlinear feedback alone can trigger global structural transitions.

References

- [1] J. Liu *et al.*, *Dynamical Phase Transition in Non-Equilibrium Networks*, Phys. Rev. Lett. 2025.
- [2] A.-L. Barabási and R. Albert, *Emergence of Scaling in Random Networks*, Science 286, 509–512 (1999).